Hacettepe University - Computer Engineering
CMP720 Embedded System Design - Spring 2026

Final Project Topic Proposal Template

(Fill in the sections below. Replace placeholders with your content.)

Group Members (Name Surname + Student ID)

Instructions: List every group member as “Name Surname — Student ID”.

- **Yusuf Tahir Orhan — N24112049**

Project Topic Title

Instructions: Provide a concise, descriptive title.

Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm for 2.5D Chiplet Networks

Project Track (choose one)

☑ **Implementation Track**

☐ **Optimization Track**

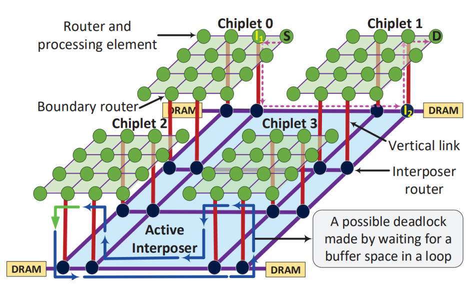
Abstract (300–500 words)

Instructions: In 300–500 words, explain: (i) the problem tackled, (ii) the proposed solution approach and methodology, (iii) high-level system architecture and major components, and (iv) an initial plan and milestones for attacking the problem. Include one crude block diagram to illustrate the idea.

Problem: 2.5D chiplet integration is a cost-effective and high-yield solution for large-scale modular systems. However, the underlying network is prone to deadlocks—even if individual chiplets are deadlock-free—and is highly susceptible to faults on the Vertical Links (VLs) that connect chiplets to the interposer[1]. Existing fault-tolerant routing techniques designed for 2D and 3D NoCs are not directly applicable to the unique constraints of 2.5D architectures[1].
Proposed Solution: This project implements DeFT (Deadlock-Free and Fault-Tolerant), a novel routing algorithm specifically designed for 2.5D integrated chiplet systems[1]. The methodology employs a Virtual Network (VN) assignment strategy to ensure deadlock-freedom and a dynamic VL selection mechanism to maintain connectivity despite hardware failures[1].
System Architecture: The target architecture consists of multiple CPU chiplets integrated on an active interposer[1]. Each chiplet is connected to the interposer through a set of bidirectional VLs[1]. The project will model this hybrid 2.5D topology where boundary routers handle the transition between local chiplet traffic and the global interposer network[1].
Methodology and Milestones:

- **Simulator Enhancement: Extending the Noxim simulator to support 2.5D topologies and VL fault injection[1].**
- **Algorithm Implementation: Developing the DeFT routing logic, including VN-assignment and fault-tolerant VL selection tables[1].**
- **Evaluation: Analyzing network reachability and latency under various fault rates[1].**

Diagram: **Proposed 2.5D Network Architecture[1].**



Motivation / Societal Impact (200–500 words)

Instructions: In 200–500 words, explain why this problem matters for the greater good (societal impact, safety, sustainability, accessibility, cost reduction, reliability, etc.). Include who benefits and how you would measure impact.

The continuous demand for higher computation power necessitates further scalability improvements in systems-on-chip (SoCs), yet conventional 2D designs suffer from high manufacturing costs and low yield[1]. While 2.5D integration offers a modular solution by placing multiple chiplets on an interposer [1], the reliability of vertical links (VLs) remains a major concern due to manufacturing defects and environmental stressors[1]. These links are highly prone to faults caused by inevitable mismatch, electromigration, and thermomigration in microbumps[1]. In standard systems, even a few VL faults can lead to catastrophic deadlocks or completely disconnect chiplets from the network[1].

The primary motivation of this project is to enhance the resilience and sustainability of modern modular processors. This project is directly aligned with my Master's thesis research, which focuses on developing reinforcement learning-based routing algorithms for 2.5D chiplet networks. By implementing the DeFT algorithm as a baseline, I aim to provide a robust deadlock-free and fault-tolerant routing solution[1]. This implementation allows the network to maintain 100% functionality even under severe fault rates, such as 25% VL failure, which significantly outperforms current state-of-the-art methods like MTR and RC.

The societal impact is reflected in reduced electronic waste and lower manufacturing costs[1]. By ensuring that chips with minor interconnect defects remain functional, we improve overall hardware sustainability[1]. Furthermore, by optimizing load distribution on VLs, DeFT reduces stress-migration-based faults and lowers power consumption by minimizing path lengths[1]. The success of this project will be measured by evaluating 100% reachability under fault scenarios and observing up to a 40% reduction in network latency during multi-application execution[1].

Related Work / References

Add a few related works from the literature, patents, or existing products to show what other researchers and developers have done in this area (3–5 works are sufficient, but you may include more). At this stage, you are not expected to conduct a full literature review; that will be done after your project idea is accepted. ***Add entries in BibTeX format to `references.bib` and cite them in your text using `\cite{<bibtexhandle>}`.*** *The References will be ordered and generated automatically. You don't need to order them inside `references.bib` or delete references that you do not cite.*

Sample citation: Real-Time Operating Systems for Embedded Systems: A Survey [1].

References

[1] E. Taheri, S. Pasricha, and M. Nikdast, "Deft: A deadlock-free and fault-tolerant routing algorithm for 2.5d chiplet networks," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pp. 1047–1052, IEEE/EDAA, 2022.

How to Submit

- Open the Google Drive sheet (submission tracker): `https://docs.google.com/spreadsheets/d/17ACM2EVBZ9tZY3XIKveWwRSbt-QfVjS7qkYcTWC3GwU/edit?usp=sharing`
- Fill in your group and project details in an appropriate row (make sure you do not delete/edit other students' entries!).
- Create a repository that you will use to submit **all** project work throughout the semester (GitHub or GitLab).
- Upload/commit your proposal files to the repository (both the `.zip` source files with `.tex` and the compiled `.pdf`).
- Paste the repository link into the corresponding column in the sheet.
- If your repository is private, add the instructor as a collaborator: `selmadilek@gmail.com`.